# SQL RANK() vs DENSE_RANK()

Both are window functions used to rank rows based on a specified order. The key difference is what happens to the rank after a tie.

## 1. Example

Suppose employees have these salaries:

| Employe | Salary |
|---|---|
| Amit | 90000 |
| Rahul | 80000 |
| Priya | 80000 |
| Neha | 70000 |
| Ravi | 60000 |

SQL query:

```
SELECT
    employee,
    salary,
    RANK() OVER (ORDER BY salary DESC) AS salary_rank,
    DENSE_RANK() OVER (ORDER BY salary DESC) AS dense_salary_rank
```

FROM employees;

Result:

| Employe | Salary | `RANK()` | `DENSE_RANK()` |
|---|---|---|---|
| Amit | 90000 | 1 | 1 |
| Rahul | 80000 | 2 | 2 |
| Priya | 80000 | 2 | 2 |
| Neha | 70000 | 4 | 3 |
| Ravi | 60000 | 5 | 4 |

## 2. Why are the results different?

### `RANK()` — leaves gaps

**1** 90k

**2** 80k

**2** 80k

**4** 70k

Both employees with 80k get rank 2. Rank 3 is skipped, so 70k gets rank 4.

**`DENSE_RANK()` — no gaps**

**1**90k

**2** 80k

**2** 80k

**3** 70k

Both employees with 80k get rank 2. The next distinct salary gets rank 3.

## 3. When should you use each?

| Requirement | Function |
| --- | --- |
| Rank students with ties and skip positions | `RANK()` |
| Rank distinct salary levels without gaps | `DENSE_RANK()` |
| Assign every row a unique sequential number | `ROW_NUMBER()` |

| | |
|---|---|
| Find the second-highest distinct salary | `DENSE_RANK()` |
| Find employees in the top 3 distinct salary levels | `DENSE_RANK()` |
| Return at most 3 employees, even if salaries tie | `ROW_NUMBER()` |

## 4. Interview example: second-highest salary

Suppose salaries are `90000, 80000, 80000, 70000`.

To find the second-highest distinct salary:

```
SELECT salary
FROM (
  SELECT
    salary,
    DENSE_RANK() OVER (ORDER BY salary DESC) AS rnk
  FROM employees
)
WHERE rnk = 2;
```

This returns `80000` for both employees if you select employee details, because both share the second-highest distinct salary.

Remember:

- `RANK()` = ties create gaps.
- `DENSE_RANK()` = ties do not create gaps.
- `ROW_NUMBER()` = every row gets a unique number.